INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 12

1. Experiment Title:

Using a Subquery

2. Aim:

Use a subquery with IN to find students who are members of both clubs, count club members, and sort club names.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

A subquery is a query nested inside another SQL query. IN checks whether a value belongs to the result set returned by the subquery. COUNT(*) counts rows.

5. Algorithm / Procedure:

1. Create club_a and club_b and insert their supplied names.
2. Use a subquery after IN to find names from club_a that also occur in club_b.
3. Count the rows in club_b and alias the result members.
4. Select all names from club_a in alphabetical order.

6. Program / Queries / Commands:

```
CREATE TABLE club_a (
    student_name VARCHAR(30)
);

INSERT INTO club_a (student_name)
VALUES ('Asha'), ('Rohan'), ('Neha');

CREATE TABLE club_b (
    student_name VARCHAR(30)
);

INSERT INTO club_b (student_name)
VALUES ('Neha'), ('Pooja'), ('Meera');

-- Task 1
SELECT student_name
FROM club_a
WHERE student_name IN (
    SELECT student_name
    FROM club_b
);

-- Task 2
SELECT COUNT(*) AS members
FROM club_b;

-- Task 3
SELECT student_name
FROM club_a
ORDER BY student_name;
```

7. Input:

club_a: Asha, Rohan, Neha. club_b: Neha, Pooja, Meera.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: student_name
Neha
Task 2: members
3
Task 3: student_name
Asha
Neha
Rohan

9. Result:

The SQL statements for Experiment 12 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is a subquery?
2. What does IN check?
3. Can a subquery be used inside WHERE?
4. What does COUNT(*) count?